IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Manuscript revised September 5, 2026.

Digital Object Identifier

PGA-UNet: A Lightweight Prompt-Guided Attention U-Net for Interactive Bone X-Ray Lesion Segmentation

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ABSTRACT Bone X-ray lesion segmentation is difficult because lesions are often small, low-contrast, and easily confused with the surrounding anatomy. In practice, the decisive step is usually localizing the suspicious region correctly, not refining the lesion's boundary. Building on this observation, we narrow the search space before segmentation through an interactive mechanism: a clinician draws a coarse bounding box around the suspicious region, and the model uses this box as a spatial guidance signal.

This article proposes PGA-UNet, a lightweight prompt-guided CNN. The bounding box is converted into a Gaussian-smoothed plateau heatmap, then used to modulate encoder features through a Prompt Spatial Gate (PSG) and to sustain its influence in the decoder through a Conditional Attention Decoder (CAD). This design does not only use the prompt to indicate where segmentation should occur, but also keeps small-lesion-relevant features active throughout the network.

We use Attention U-Net without a prompt as the automatic reference model, and SAM-Med2D, MedSAM, and ScribblePrompt as prompt-matched references. Experiments on the BTXRD and FracAtlas datasets, at three resolutions (128, 256, and 512), under two box conditions, covering and controlled off-center, show that prompt-free segmentation is a very hard problem: Attention U-Net reaches only 0.52 and 0.34 Dice on the two datasets. Given the same box, PGA-UNet matches or outperforms all three fine-tuned promptable foundation models and two conventional prompt-matched baselines, with a clear advantage under off-center prompts. The margin is widest against SAM-Med2D and on the smallest-lesion group: fine-tuned SAM-Med2D drops to 0.26 to 0.37 Dice there, while PGA-UNet still holds 0.65 to 0.72. The results are stable across repeated random data splits, and PGA-UNet is about 92 times lighter than SAM-Med2D in parameter count. The system also exposes two label-free support signals computed from the model's output, a self-assessment score for mask reliability and a short candidate-region shortlist, both preliminary.

It should be noted that every box in the experiments is simulated around an already-annotated lesion, so the results above only reflect segmentation ability once a good prompt is already available. This article does not answer questions about: fully automatic lesion detection, partial-coverage or off-target boxes, calibrated confidence, real clinical usability, patient-level generalization, or transfer to another dataset.

INDEX TERMS bone X-ray, interactive segmentation, medical image segmentation, prompt-guided learning, Attention U-Net

I. INTRODUCTION

THANKS to its low cost, speed, and accessibility, bone X-ray imaging is widely used in screening, diagnostic support, and treatment monitoring. However, segmenting lesions in this type of image is not simple when the lesion is very small, low-contrast, or partially obscured by the surrounding bone structure. In such cases, the main difficulty lies not in drawing an accurate boundary, but in localization: the model must find the suspicious region in the whole image before it can refine any boundary.

This is why small-lesion segmentation is rarely just a boundary-drawing problem. When a lesion occupies only a tiny fraction of the image, searching the whole image is both costly and error-prone. The natural approach is to narrow the search space first, then segment within the narrowed region. Fully automatic segmentation methods, typified by U-Net and

Attention U-Net [3], [4], try to learn both localization and segmentation implicitly within the same network, with no external help at all.

Another approach is to draw on external guidance about the lesion's location. In clinical practice, a clinician can easily and quickly draw a coarse bounding box around a suspicious region, something an automatic model struggles to do well for very small lesions. Seen this way, the bounding box is not merely an interactive convenience, but a practical way to inject spatial knowledge into the model before detailed segmentation begins. Many prompt-based interactive segmentation methods have grown out of this idea, from simple operations such as clicks and scribbles [5], [6] to recent foundation models such as SAM and SAM-Med2D [1], [2].

Even so, one question remains unanswered: once the prompt has identified the region of interest, should it only act as a positional constraint for generating the mask, or should it continue to shape feature extraction, so that an already faint lesion signal is not lost through the downsampling layers? This question matters most for very small lesions, where even a small spatial deviation can substantially affect the final outcome.

This article answers that question with PGA-UNet, a lightweight prompt-guided CNN in which the bounding box is treated as a structured spatial prior, not merely an auxiliary input channel. The box is encoded into a Gaussian-smoothed plateau heatmap, used to modulate encoder features through a Prompt Spatial Gate (PSG), and then reused in the decoder through a Conditional Attention Decoder (CAD). As a result, prompt-guided lesion features stay active throughout the network, rather than being confined to limiting segmentation within the given box.

This perspective also determines how we design and read the experiments. Attention U-Net receives no spatial guidance from a clinician, so its role is only to show how hard the problem becomes when localization must be fully automatic. Promptable foundation models (SAM-Med2D, MedSAM, ScribblePrompt), by contrast, receive the same bounding-box input and serve as the direct comparison. The central question of this article is therefore not whether a prompt is useful in general, but whether a lightweight architecture can exploit a clinician's coarse prompt more effectively, by preserving weak lesion signals instead of treating the prompt merely as a positional constraint.

The main contributions of this article are as follows.

- We propose PGA-UNet: a lightweight, prompt-guided architecture for interactive bone X-ray lesion segmentation.
- We build a prompt-conditioning mechanism with three parts: a Gaussian-plateau prompt, an encoder-side Prompt Spatial Gate (PSG), and a decoder-side Conditional Attention Decoder (CAD).
- We build an offline training and online interactive inference framework, together with explicit mathematical definitions for prompt encoding, encoder-side gating, and decoder-side conditioning.
- We evaluate the method on the BTXRD and FracAtlas datasets under covering and off-center prompt conditions, comparing against an automatic Attention U-Net, two Attention U-Net variants that we add ourselves to see how a conventional network behaves when it is also given the same box prompt (one that concatenates the prompt as an input channel, one that only looks at the prompt-cropped region), and three fine-tuned promptable models (SAM-Med2D, MedSAM, ScribblePrompt); using Dice and IoU for region overlap, CBL for center-based localization, and HD95 for boundary distance, together with Monte Carlo cross-validation and computational efficiency measurements.
- Through a small-lesion analysis and an Attention U-Net-defined difficulty split, we show that the benefit of prompt-conditioned processing is concentrated exactly where automatic localization fails and where lesions are smallest.

Beyond these contributions, the system also exposes two label-free support signals, computed directly from the trained model's output: a self-assessment score that ranks mask reliability, and a short candidate-region shortlist. Both are presented as preliminary and are not central contributions of this article.

II. RELATED WORK

For a long time, small-lesion segmentation in medical imaging has run into a simple fact: to segment a very small target well, a model must first narrow down where to look. In the fully automatic direction, this narrowing is learned implicitly inside the network. U-Net is the multi-scale encoder-decoder architecture that laid the foundation for biomedical segmentation [3], and Attention U-Net later added attention gates on the skip connections so the decoder could filter out irrelevant spatial responses more selectively [4]. Self-configuring pipelines such as nnU-Net go further still, automatically adapting preprocessing, network shape, and training schedule to each dataset [10]. These models raise automatic segmentation quality considerably, but still rest on a shared assumption: the network must localize and draw the mask from the input image alone, with no hint at all about where the lesion lies.

In parallel, interactive medical segmentation has also developed, starting from the observation that even a small amount of user guidance can substantially simplify the problem. Early methods used clicks, scribbles, or geodesic cues [5], [6]; later, bounding boxes were added as a fast, coarse form of prompt. More recently, interactive models such as ScribblePrompt can handle several prompt types within a single framework, across many biomedical datasets [11]. Broadly, these methods shift part of the localization burden from the model to the user. For very small X-ray lesions, that shift is especially valuable, since a clinician can usually point out the suspicious region coarsely even when an automatic system struggles to find it reliably in the whole image.

TABLE 1. Representative prior work and the remaining gap addressed in this article.

Work	Principle	Method	Performance Evidence	Pros	Cons
U-Net [3]	Fully convolutional encoder-decoder	Multi-scale skip-connected segmentation network	Widely used as a canonical biomedical segmentation baseline	Simple, efficient, strong reference model	No prompt guidance; all localization must be automatic
Attention U-Net [4]	Skip-feature filtering by attention gates	U-Net with gated skip connections	Medical segmentation tasks reported with overlap metrics such as Dice and IoU	Stronger automatic baseline than plain U-Net	Attention is still learned without explicit prompt conditioning
nnU-Net [10]	Self-configuring automatic pipeline	Dataset-adaptive preprocessing, architecture, and training	Strong automatic results across many medical benchmarks	Robust automatic pipeline with little manual tuning	No prompt mechanism; localization remains fully automatic
DeepIGeoS and related interactive methods [6]	User guidance reduces search space	Geodesic or interaction-aware segmentation	Interactive medical segmentation under user input	Better reflects clinical interaction	Interaction mechanism can be task-specific or costly to maintain
ScribblePrompt [11]	General interactive biomedical segmentation	One model for scribbles, clicks, and boxes across datasets	Multi-dataset interactive evaluation with several prompt types	Flexible, realistic interaction types	General-purpose, not a lightweight radiograph-specific prompt prior
SAM-Med2D / MedSAM [2], [9]	Promptable foundation segmentation	Large pretrained model with prompt-driven mask decoding	Multi-dataset medical evaluation with prompt-based metrics	Flexible and broadly applicable	Heavy; prompt influence is not designed as a lightweight end-to-end radiograph-specific prior
EMedSAM / Med-SA [12], [13]	Adapter tuning of SAM for medical images	Lightweight adapters on a frozen SAM backbone	Medical segmentation with reduced adaptation cost	Cheaper specialization of a foundation model	Still a large SAM backbone; prompt acts mainly at the mask decoder
This work	Prompt as a dense spatial prior throughout the network	Gaussian plateau prompt + PSG + CAD in a lightweight CNN	BTXRD and FracAtlas; Dice, CBL, HD95, efficiency, and small-lesion analysis	Lightweight, prompt-aware from encoder to decoder, evaluated under controlled prompt displacement	Still requires supervised masks and fixed-resolution preprocessing

Promptable foundation models push this idea further. SAM pioneered turning prompt-driven segmentation into a general-purpose task [1]; SAM-Med2D and MedSAM brought it into medicine [2], [9]; adapter-based variants such as EMedSAM and Med-SA reduce the cost of specializing these models to medical data [12], [13]. These models are quite flexible, but they consume the prompt mainly at the mask decoder, carry a very large backbone, and none is tuned specifically for a single radiograph task. For very small bone lesions, this is exactly the gap: the prompt only marks where the mask goes, while no mechanism continues to shape feature extraction afterward, exactly where a weak signal is easily lost.

This article starts from that very gap. On grayscale bone radiographs, knowing roughly where the lesion is only solves half the problem; the harder part is holding on to faint evidence while still filtering out noise, after the search space has been narrowed. We argue that a lightweight prompt-guided CNN can do this better if the prompt is kept continuously active from the first encoder stage to the last decoder stage. In this article we study that setting directly, using SAM-Med2D, MedSAM, and ScribblePrompt as the promptable reference models.

III. METHOD

A. OFFLINE AND ONLINE FRAMEWORK

The system operates in two stages. In the offline stage, the model is trained on X-ray images that already have lesion polygons. Each polygon is turned into a training prompt by generating a simulated bounding box, then converting that box into a Gaussian-smoothed plateau heatmap. The network learns to map the (image, prompt) pair to a binary lesion mask. In the online stage, the user only needs to draw a coarse bounding box around the suspicious region on a new image; the same prompt-construction procedure is applied again, and the trained model produces the final predicted mask.

What sets this apart is where the prompt lives: instead of entering only once at the first layer, it is kept as a dense spatial prior, active throughout the encoder and decoder, which should help when the box is coarse, slightly off-center, or wrapped tightly around a very small lesion.

An interactive demonstration of the online stage is given in Figure 1: the user draws a box on the X-ray image, and the model returns the mask together with CAD's prompt-use score and the self-assessment score of Section III-F, both of which require no ground truth.

B. PROBLEM SETTING

Given a grayscale radiograph $\mathbf{I} \in \mathbb{R}^{H \times W}$ and a user-provided bounding box $\mathbf{B}$, the goal is to predict a binary lesion mask $\hat{\mathbf{M}}$. Let $\mathbf{H}(\mathbf{B})$ be the prompt heatmap derived from the box. PGA-UNet predicts

$$\hat{\mathbf{M}} = \mathbf{1}\left[f_{\mathrm{PGA}}\left(\mathbf{I}, \mathbf{H}(\mathbf{B}); \theta\right) \geq 0.5\right]. \tag{1}$$

C. PROMPT REPRESENTATION

Instead of encoding the input prompt as a hard binary rectangle, we rasterize the interior of the box into a plateau map at the original image resolution, soften its edges with a fixed 31×31 Gaussian kernel, and only then resize and pad this heatmap down to the target square resolution together with the image. The kernel size does not change with the target resolution (256×256 or 512×512): applying it once, at the original resolution, before downsampling, keeps the effective blur consistent regardless of which square frame the network ultimately sees. The plateau covers exactly the area of the box (expanded if applicable), with no additional minimum margin added; coverage of the region of interest is already guaranteed by the box-expansion step, so no separate margin step is needed.

At evaluation time, the prompt is tested under two fixed conditions. The covering condition (`center_zoom`) scales the tight lesion box outward from its own center by a fixed factor of 3.0. The off-center condition (`center_shift`) applies the same scaling, then shifts the box by a fraction of the tight-box size (capped by `shift_ratio`=0.5), expanding it back if needed so the box still fully contains the lesion. At test time, this shift is fixed per lesion (seeded from the sample index), so every model is evaluated on exactly the same set of off-center boxes; during training, the shift is redrawn randomly at each iteration. For each dataset and resolution, we train a single model using `center_mixed`: `center_shift` is chosen with probability 0.8 and `center_zoom` with probability 0.2, while at test time the two conditions are scored separately. The task scope assumes the user has already identified the suspicious region and deliberately draws a box that covers the lesion; partial-coverage boxes, negative boxes, and off-target boxes all lie outside this controlled protocol, and whether a clinician can find the correct region in the first place is a separate question, outside the scope of this article.

Let (x_1, y_1, x_2, y_2) denote the tight lesion box, with $w = x_2 - x_1$ and $h = y_2 - y_1$. The simulated covering box uses

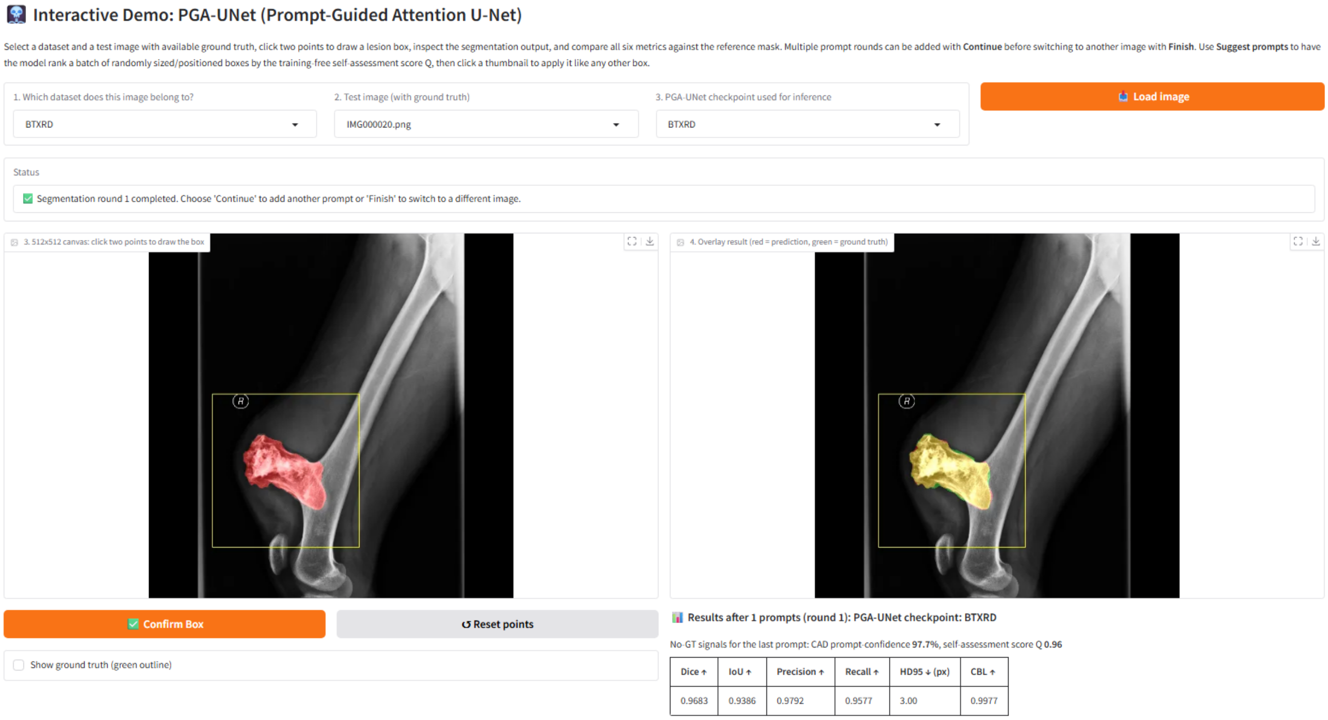


FIGURE 1. Interactive demonstration of the online stage on a BTXRD case. The clinician draws a box on the left canvas; the right panel overlays the predicted mask, and the panel below reports the reference metrics together with the no-ground-truth CAD prompt-use score and the self-assessment score *Q*. A FracAtlas case is shown in the same layout in the supplementary material.

exactly the fixed center-scaling rule above, applied during both training and evaluation. The off-center box reuses that scaled box, then applies the deterministic test-time shift rule implemented in `dataset.py`, while still preserving lesion coverage. The scale factor, shift ratio, and Gaussian kernel size were all chosen from preliminary experiments on the training and validation data and then locked; the test split played no part whatsoever in choosing the protocol or any hyperparameter. These are simply fixed protocol values used throughout the article, not a globally optimal configuration. This simulation naturally cannot substitute for a prompt actually drawn by a radiologist, but it offers a controlled way to measure the model's sensitivity to a predefined prompt displacement.

In summary, prompt construction consists of three steps: (1) derive the lesion box from the annotated polygon or from user input, then expand it as described; (2) rasterize the expanded region into a plateau mask at the original image resolution; (3) smooth the plateau's edge with the fixed Gaussian kernel before resizing to the target resolution. This avoids the hard-edge dependence of a binary box while still keeping a clear spatial prior.

D. PROMPT SPATIAL GATE

Let $\mathbf{x}^l$ denote the encoder feature map at level l. PSG modulates it with a prompt-derived attention map $\mathbf{A}^l_{\mathrm{PSG}}$:

$$\tilde{\mathbf{x}}^l = \mathbf{x}^l \odot \left(1 + \alpha^l_{\mathrm{PSG}} \mathbf{A}^l_{\mathrm{PSG}}\right), \tag{2}$$

where $\odot$ denotes element-wise multiplication, and α^l_{PSG} is a learnable parameter, initialized at 0.1 and clamped to $[0, 1]$. The residual form preserves the original feature stream while emphasizing prompt-relevant regions.

E. CONDITIONAL ATTENTION DECODER

CAD extends decoder attention by conditioning the skip gate on prompt-encoded features. Let $\mathbf{g}^l$ be the decoder gating feature and $\mathbf{p}^l_{\mathrm{enc}}$ the prompt encoding at the same scale. The conditioned gate is

$$\mathbf{g}'^{\,l} = \mathbf{g}^l + c^l \alpha^l_{\mathrm{CAD}} w_l \mathbf{p}^l_{\mathrm{enc}}, \tag{3}$$

where c^l is a prompt-use coefficient derived from the decoder, α^l_{CAD} is learnable, and w_l is a fixed depth coefficient. As a result, the gate stays responsive to the prompt even when the box is not perfectly centered.

The prompt encoder inside CAD consists of two 3×3 convolutional layers with instance normalization and ReLU. The prompt-use coefficient c^l is produced by global average pooling, followed by a 1×1 convolution and a sigmoid; this coefficient is computed only inside the gate, used to scale the prompt term at that level, and is not exposed or analyzed as a separate model output in this article. The decoder's depth coefficients are fixed at $\{1.0, 0.7, 0.4, 0.2\}$ from deep to shallow. With a feature scale of 4, the channel widths are $\{16, 32, 64, 128, 256\}$ from the shallowest encoder stage to the bottleneck, and the whole model has about 2.95 million parameters. These protocol values were all chosen from pre-

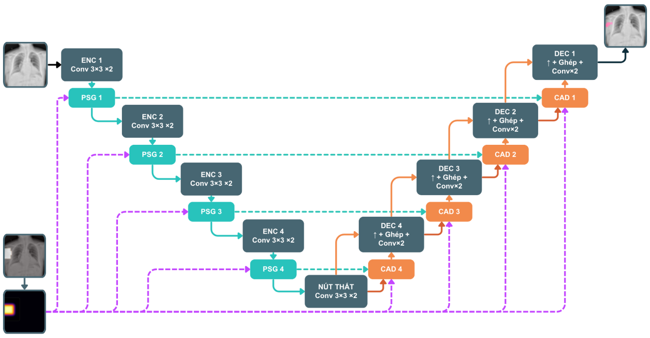


FIGURE 2. **Overview of PGA-UNet. Prompt information derived from the bounding box is injected into encoder features through PSG and reused in decoder skip attention through CAD.**

liminary experiments on the training and validation sets, held fixed throughout the main evaluation, with the test split never involved in tuning; we make no claim of optimality. The CAD mixing coefficient is parameterized through a sigmoid and initialized near 0.3.

In summary, the method consists of four steps: (1) preprocess the image and prompt to the same square resolution; (2) encode the prompt-relevant spatial prior with PSG in the encoder; (3) propagate prompt-conditioned context through CAD in the decoder; (4) predict the final binary lesion mask.

F. TRAINING-FREE SELF-ASSESSMENT SCORE

Besides the predicted mask, PGA-UNet also returns a scalar Q, estimating how trustworthy a prediction is without needing the true answer. Q is not a learned quantity; it is computed only from the model's own output at inference time, as the average of two cues. Q is used as a support signal, letting the user look more critically at a given mask, not as a core component of the method.

The first cue, S_{sharp}, measures how decisive the predicted mask is. Let p be the sigmoid probability map and $\Omega = \{p > 0.5\}$ the predicted positive region. Then

$$S_{\text{sharp}} = \text{clip}_{[0,1]}\left(\frac{1}{|\Omega|}\sum_{u\in\Omega}\big(2\,p(u)-1\big)\right). \qquad (4)$$

A mask with high confidence, i.e., probabilities close to 1, scores close to 1; conversely, a mask with many pixels sitting near the decision threshold scores lower.

The second cue, S_{stab}, measures how stable the prediction is under mild prompt perturbation. The box is randomly nudged K times, each time with a small shift and scale change; the model is rerun for each nudged box, and S_{stab} is exactly the average IoU between the original binary mask $\hat{\mathbf{M}}$ and the masks obtained from those nudges, $\hat{\mathbf{M}}_k$:

$$S_{\text{stab}} = \frac{1}{K}\sum_{k=1}^{K}\text{IoU}\left(\hat{\mathbf{M}}, \hat{\mathbf{M}}_k\right). \qquad (5)$$

TABLE 2. **Image-level train / validation / test splits. Each lesion polygon is one promptable sample.**

Dataset	Images			Lesion polygons		
	Train	Val	Test	Train	Val	Test
BTXRD	1,493	187	187	1,848	238	232
FracAtlas	573	72	72	730	95	92

A prediction that stays nearly unchanged when the box is nudged slightly is considered more trustworthy than one that collapses or drifts elsewhere.

The final score is $Q = \frac{1}{2}\left(S_{\text{sharp}} + S_{\text{stab}}\right)$, with $K = 6$, a maximum shift of 0.12 of the box side, and a scale factor within ± 0.10. It should be emphasized that Q is only a heuristic ranking signal, not a calibrated probability: a value of 0.7 does not mean a 70% probability that the mask is correct. Q also presupposes that a lesion lies inside the box: it measures mask quality, not lesion presence.

We reuse this same score to shortlist candidate regions when no prompt is given: a set of boxes with varying size and position is sampled, each box is scored with Q, and the highest-scoring few are returned for the clinician to accept, adjust, or discard. This is only a scaffold for a small suggestion feature we are working toward, not an automatic detector.

IV. EXPERIMENTAL SETUP

A. DATASETS

We run our experiments on two public bone X-ray datasets. BTXRD is a primary bone tumor dataset with polygon annotations supporting classification, localization, and segmentation [7]. FracAtlas is a fracture dataset with the same label types [8]; we convert its segmentation annotations to the same polygon format used for BTXRD before use, and the conversion is released with the reproducibility package. Both datasets are split at the image level, so that every polygon belonging to the same image always falls in the same split (Table 2).

Because the public metadata do not provide enough information to verify strict patient-level separation, images from the same patient could in principle still fall into different splits; we regard this as a limitation of this evaluation. In image-level evaluation, if an image has multiple lesions, the model is run separately for each box, and the resulting probability maps are merged by a pixelwise maximum before thresholding at 0.5; the ground-truth masks are likewise merged by union.

Input images are resized isotropically to preserve aspect ratio, then padded to a square resolution of 256×256 or 512×512. Images, masks, and prompt maps all undergo this same geometric preprocessing.

B. TRAINING DETAILS

All CNN models were implemented in PyTorch and trained on a single NVIDIA Tesla T4 GPU with 16 GB memory. PGA-UNet and Attention U-Net were trained separately for each dataset and each resolution; in other words, this article studies one shared architecture family instantiated as separate models, with separate parameter sets for BTXRD and FracAtlas, rather than training a single joint model on both. We used the AdamW optimizer with learning rate 10^{-4} and weight decay 10^{-4}; the loss was the sum of binary cross-entropy and Dice loss. Because the segmentation targets are often small, we additionally tested training PGA-UNet at 512×512 with the Dice term replaced by a size-conditioned Tversky loss, and in a separate run, by a Focal Dice loss, with every other setting held fixed. These two replacements are mutually exclusive and are reported later only as a negative control on the training objective. Batch size was 4, training ran for at most 150 epochs, with early stopping at patience 15. All main and ablation experiments share the same fixed training seed, 22120196, which seeds Python, NumPy, and PyTorch on both CPU and CUDA. The primary model-selection criterion was image-level merged Dice, measured on the validation set under the `center_shift` condition.

During training, we applied augmentation consisting of horizontal flipping and random rotation within $\pm 15°$. PGA-UNet is trained with `center_mixed`: `center_shift` is chosen with probability 0.8 and `center_zoom` with probability 0.2. Because the prompts are all simulated from labeled lesion regions, this is a prompted-segmentation protocol under controlled conditions; the article should be read accordingly, namely as a study of simulated lesion-covering boxes around user-identified lesions, not as an evaluation of unconstrained lesion detection or of boxes fully drawn by a clinician in real practice.

For the fine-tuned SAM-Med2D comparison, the pre-trained image encoder was frozen, with only its lightweight Adapter layers trained further, while the prompt encoder and mask decoder were fully updated on each dataset, under the same split protocol. SAM-Med2D was fine-tuned with box prompts only, using the same center-scaled covering and off-center protocol as PGA-UNet; the point-prompt branch and iterative point refinement were both disabled. This box construction differs from the small random-jitter `get_boxes_from_mask` sampling used by the original SAM-Med2D authors, which we did not apply here. Both validation during SAM-Med2D fine-tuning and the final test-split evaluation report both prompt conditions separately; batch size was 4, at most 150 epochs, early stopping patience 30, and learning rate 10^{-5}. This matched protocol reduces differences caused by prompt distribution and resolution, but cannot eliminate differences in model scale, initialization, or optimization.

We additionally ran stability experiments using Monte Carlo cross-validation, that is, repeated random image-level splits, as opposed to strict non-overlapping k-fold partitioning. These experiments keep the training seed fixed at 22120196 and vary only the split seed, across the values 1, 2, 3, and 4.

Two details are worth noting when interpreting the evaluation results. First, prompt construction itself does not depend on resolution: the plateau heatmap is built and smoothed with a fixed 31×31 Gaussian kernel at the original image resolution, and only then resized and padded down to 256×256 or 512×512 together with the image. In other words, the same absolute blur is compressed differently at the two target resolutions, rather than being parameterized separately for each. Second, the Monte Carlo experiments are only auxiliary stability evidence, not intended to replace the fixed train/validation/test split used throughout the main results tables.

C. BASELINES AND METRICS

We compare PGA-UNet against an automatic baseline, Attention U-Net [4], and three fine-tuned promptable baselines: SAM-Med2D [2], MedSAM [9], and ScribblePrompt-UNet [11]. Attention U-Net is trained without a prompt, to show how hard the task becomes when localization must be fully automatic. Among the promptable baselines, SAM-Med2D is fine-tuned and evaluated at 256×256 using the same dataset splits and the same box prompts as PGA-UNet, so PGA-UNet at 256 against SAM-Med2D at 256 is the article's resolution-matched prompt comparison. MedSAM and ScribblePrompt-UNet run at their own native resolutions (1024 and 128); Table 6 therefore also lists PGA-UNet at 512 as its best-performing configuration and compares the frame-invariant metrics (Dice, IoU, CBL) across all models.

To isolate whether PGA-UNet's advantage comes from its architecture or simply from having access to a box prompt at all, we add two conventional prompt-matched baselines of our own, both built on the same Attention U-Net backbone and given exactly the same box prompt that PGA-UNet receives, but without PGA-UNet's Gaussian-prior heatmap, Prompt Spatial Gate, or Conditional Attention Decoder. The first baseline, Attention U-Net with a prompt channel, concatenates the prompt heatmap as a second input channel and predicts on the full image through a plain skip-connection decoder. The second baseline, Attention U-Net on prompt crops, narrows the network's field of view: the input image is cropped to the prompt box before resizing to the model's input resolution, the plain Attention U-Net predicts a mask on that crop alone, and the result is pasted back into the full frame for evaluation. Both baselines are trained under the same covering and off-center box protocol as PGA-UNet, so any remaining gap reflects a difference in architecture rather than an unfair prompt. When comparing models against each other, we use the pasted-back full-frame metric for the prompt-crop baseline, not the diagnostic value measured inside the crop frame.

Every comparison table in this article reports Dice and IoU for region overlap, CBL for localization, and HD95 for boundary distance. Dice and IoU are fairly strongly correlated overlap measures; we place them side by side only for the reader's convenience in cross-checking. On the small-lesion subsets specifically, IoU has a low absolute value and is quite sensitive to a few boundary pixels, so Dice is the primary overlap metric used there. Let (x_p, y_p) be the predicted mask centroid, (x_g, y_g) the ground-truth centroid, and D_{GT} the diagonal length of the ground-truth bounding box; CBL is defined as follows:

$$\mathrm{CBL} = \max\left(0, 1 - \frac{\sqrt{(x_p - x_g)^2 + (y_p - y_g)^2}}{D_{\mathrm{GT}}}\right). \quad (6)$$

HD95 is computed from bidirectional boundary distances. When both masks are empty, HD95 is set to 0; if only one mask is empty, HD95 is set to the input side length S (256 or 512 pixels). No sample is discarded from the computation. For comparisons spanning resolutions, we additionally report HD95 in normalized form, $\mathrm{HD95}/(\sqrt{2}\,S)$, since a raw pixel distance cannot be directly compared across the 128, 256, and 512 frames. Some subgroup comparisons mix models trained at 256×256 and 512×512; in that case, everything is scored together in a shared 512×512 frame, with the 256×256 predictions upsampled before scoring, so every model is measured against exactly the same ground-truth mask, rather than in its own native frame. The promptable-baseline tables (Tables 6 and 10) are the exception: each model there is scored in its own native frame (PGA-UNet at 256 or 512, SAM-Med2D at 256, MedSAM and ScribblePrompt-UNet at the original image resolution), so their raw HD95 values are shown for completeness but are not comparable across model families and are left unbolded. Dice, IoU and CBL are frame-invariant and remain the basis for comparison there.

For the training-free self-assessment score, each lesion prompt yields a score Q together with the true thresholded Dice of its own predicted mask; we report the Pearson and Spearman correlation between the two, separately for the covering and off-center conditions. For the candidate-region shortlist, each test image is sampled with 50 boxes whose side length is uniform in $[0.15, 0.55]$ of the frame, each box is scored by Q, and the five highest-scoring boxes are kept; each retained box is then scored by coverage, the fraction of ground-truth lesion pixels it contains, and no predicted mask enters this coverage measure. As a negative control, the same 50-box sampling is also run on the lesion-free images of each dataset (1879 for BTXRD, 3366 for FracAtlas); with no lesion present there is no coverage to compute, so we report only the distribution of Q.

D. STATISTICAL ANALYSIS

The main comparison tables (Tables 4, 6, 5) report point estimates on the fixed test split; we did not retrain every baseline across multiple splits, so those tables carry no paired tests. The stability of PGA-UNet itself across splits is quantified by the Monte Carlo cross-validation of Section V-H (Dice standard deviation ≤ 0.011). Where per-image metrics are available, namely the ablation (Section V-G) and the self-assessment analysis (Section V-L), we test differences with a paired sign-flip randomization test on per-image Dice (20,000 permutations, two-sided) and report a percentile bootstrap 95% confidence interval for the mean difference (20,000 resamples); for the self-assessment correlations we report a permutation p-value and a bootstrap 95% interval for Spearman's ρ. All resampling uses the fixed seed 22120196.

TABLE 3. PGA-UNet across resolutions, image-level merged. Dice, IoU, CBL and HD95 (native px and normalized HD95/($\sqrt{2}\,S$)) given as covering / off-center.

Dataset	Res.	Dice	IoU	$\mathrm{HD95}_{\mathrm{px}}$	$\mathrm{HD95}_n$	CBL
BTXRD	128	0.714 / 0.717	0.572 / 0.577	5.8 / 5.3	0.032 / 0.029	0.859 / 0.857
	256	0.762 / 0.743	0.633 / 0.611	10.9 / 12.8	0.030 / 0.035	0.902 / 0.885
	512	**0.782 / 0.774**	**0.661 / 0.652**	22.8 / 24.5	0.031 / 0.034	**0.918 / 0.912**
FracAtlas	128	0.596 / 0.542	0.445 / 0.389	8.1 / 9.5	0.044 / 0.053	0.849 / 0.790
	256	0.629 / 0.594	0.470 / 0.436	10.6 / 11.7	0.029 / 0.032	0.895 / 0.849
	512	**0.727 / 0.713**	**0.585 / 0.570**	10.5 / 10.4	**0.015 / 0.014**	**0.913 / 0.897**

E. SCOPE OF ANALYSIS

This article focuses only on prompt-guided segmentation itself. A lesion-screening classification stage was explored in a separate study and is not part of this article. The self-assessment score and the candidate-region shortlist are treated only as auxiliary analyses, within the same narrow scope already stated in Section III-F; apart from one negative control on lesion-free images, they are not extended any further here. To keep the article compact, we retain only the subgroup analyses that most directly support the main claim: the role of prompt-guided localization, sensitivity to controlled prompt displacement as defined above, and effectiveness on very small lesions. For each dataset, the small-lesion subgroup is defined as the 50 test images with the smallest total lesion area. In the three-resolution small-lesion comparison, this same set of images is held fixed and evaluated in turn at 128, 256, and 512; predictions and ground truth are merged at image level at each resolution, HD95 is reported both in native pixels and in the normalized form, and the degradation is measured relative to the 512 result. It should also be reiterated: partial-coverage boxes, negative boxes, and off-target boxes are outside the scope of this study, so the article's claims do not rely on them.

V. RESULTS

Unless stated otherwise, every table reports current-protocol numbers: `center_mixed` training with 80% `center_shift` and 20% `center_zoom`, 150 epochs, model selection by image-level merged validation Dice under `center_shift`, and image-level merged test evaluation. The two test prompt conditions, covering (`center_zoom`) and off-center (`center_shift`), are reported separately whenever a table carries both.

A. MAIN RESULTS ACROSS DATASETS AND RESOLUTIONS

PGA-UNet was trained from scratch at 128×128, 256×256, and 512×512 on each dataset, giving six models. Table 3 reports them under both prompt conditions.

Higher resolution helps at every step: Dice consistently follows $512 > 256 > 128$ on both datasets and under

TABLE 4. Comparison at 512 × 512 under covering prompts. Attention U-Net (no prompt) receives no box; the other rows use the same box PGA-UNet receives. ´Dice is each row minus PGA-UNet.

Dataset	Model	Dice	ΔDice	IoU	HD95	CBL
BTXRD	Attention U-Net (no prompt)	0.517	−0.265	0.418	120.4	0.620
	Attention U-Net + prompt channel	0.760	−0.022	0.635	33.0	0.895
	Attention U-Net + prompt crop	0.741	−0.041	0.615	24.7	0.912
	PGA-UNet	**0.782**	—	**0.661**	**22.8**	**0.918**
FracAtlas	Attention U-Net (no prompt)	0.342	−0.385	0.253	139.9	0.478
	Attention U-Net + prompt channel	0.593	−0.134	0.434	22.9	0.887
	Attention U-Net + prompt crop	0.590	−0.137	0.451	22.0	0.865
	PGA-UNet	**0.727**	—	**0.585**	**10.5**	**0.913**

both prompt conditions, with FracAtlas gaining more, which is reasonable since its inherently thin fracture lesions lose proportionally more detail as the image shrinks. Because HD95 in raw pixels cannot be compared across differently sized frames, we also report the normalized form: it stays near 0.03 across the BTXRD resolutions and decreases on FracAtlas as the frame grows, showing that boundary error does not worsen at higher resolution. Note that each model is evaluated only on the dataset it was trained on; the article makes no claim about cross-dataset transfer.

B. COMPARISON AGAINST AUTOMATIC AND PROMPT-MATCHED ATTENTION U-NET

With no support at all in finding the lesion, Attention U-Net reaches only 0.517 Dice on BTXRD and 0.342 on FracAtlas at 512 × 512, with HD95 exceeding 120 pixels on both datasets (Table 4). This number should not be read as a verdict on the architecture, since the model itself has never seen the box; it only shows how hard automatic localization is, exactly the part of the job that a prompt helps remove.

Given the same box, the plain Attention U-Net does considerably better, but a gap to PGA-UNet remains (Table 4, ΔDice column). PGA-UNet surpasses both prompt-matched variants on both datasets, with a small margin on BTXRD but a much wider one on FracAtlas, and lower HD95 than either.

Localization therefore accounts for most of the difficulty here, but the box alone does not fully explain the results. The way PGA-UNet exploits that box still produces an additional, consistent margin over a conventional network given the same input.

C. ATTENTION U-NET TOP-DICE AND BOTTOM-DICE SUBSETS

To see where that margin comes from, we ranked the test images by Attention U-Net Dice, kept the top 50 and bottom 50 per dataset, and re-ran all four models on those exact images (Table 5, covering prompts). On the top-50, the prompt-guided models perform quite similarly. In contrast, on the bottom-50, Attention U-Net collapses outright, down to 0.031 on BTXRD and 0.169 on FracAtlas, while PGA-UNet still holds 0.712 and 0.687. The two conventional prompt baselines also recover somewhat, but less: on the bottom group PGA-UNet still leads them by 0.04 to 0.07 Dice on BTXRD and by 0.10 to 0.19 on FracAtlas. This shows that having a prompt is necessary but not sufficient; the benefit of

TABLE 5. Top-Dice and bottom-Dice subsets at 512 × 512, 50 images per group, covering prompts. Groups are defined by plain Attention U-Net.

Dataset	Group	Model	Dice	IoU	HD95	CBL
BTXRD	Top-50	Attention U-Net	0.872	0.775	35.0	0.945
		AttUNet + channel	0.846	0.740	24.3	0.942
		AttUNet + crop	0.844	0.739	27.5	0.946
		PGA-UNet	0.862	0.763	19.2	0.944
	Bottom-50	Attention U-Net	0.031	0.016	277.8	0.105
		AttUNet + channel	0.668	0.541	43.3	0.831
		AttUNet + crop	0.652	0.518	17.6	0.875
		PGA-UNet	**0.712**	**0.589**	23.2	**0.879**
FracAtlas	Top-50	Attention U-Net	0.493	0.365	85.7	0.640
		AttUNet + channel	0.605	0.447	26.6	0.887
		AttUNet + crop	0.658	0.515	24.9	0.881
		PGA-UNet	**0.754**	**0.615**	10.6	**0.927**
	Bottom-50	Attention U-Net	0.169	0.106	182.3	0.315
		AttUNet + channel	0.574	0.416	25.3	0.875
		AttUNet + crop	0.509	0.369	26.0	0.834
		PGA-UNet	**0.687**	**0.535**	11.8	**0.896**

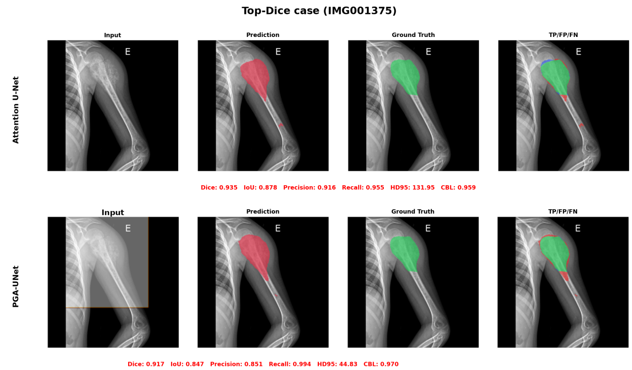


FIGURE 3. Attention U-Net (no prompt) vs PGA-UNet on BTXRD, Attention U-Net top-Dice case (IMG001375). Both models do well on an easy case.

prompt-conditioned processing is concentrated exactly where automatic localization fails. It should also be noted that these two groups are defined entirely by Attention U-Net, not by any other objective difficulty criterion.

D. COMPARISON AGAINST PROMPTABLE BASELINES

Three promptable foundation models receive the same box: SAM-Med2D (brought into medicine by broad fine-tuning), MedSAM (medical fine-tuning, box-only), and ScribblePrompt-UNet (a lightweight UNet, not a SAM-style architecture). In zero-shot mode, all three are close to unusable on bone X-ray images: 0.25 to 0.43 Dice, with very low precision (0.16 to 0.34) because the mask spills outside the lesion. Fine-tuning on the same split and the same prompt protocol adds 0.24 to 0.42 Dice (Table 6).

At the resolution-matched setting, PGA-UNet at 256 leads fine-tuned SAM-Med2D at 256 by 0.132 Dice on BTXRD (0.762 vs 0.630) and by 0.066 on FracAtlas (0.629 vs 0.563), with CBL also higher (0.902 vs 0.836; 0.895 vs 0.846).

MedSAM and ScribblePrompt-UNet run at their native 1024 and 128, so we compare the frame-invariant metrics. PGA-UNet at its best 512 configuration (0.782 on BTXRD) leads both fine-tuned MedSAM (0.724) and ScribblePrompt-UNet (0.653). On FracAtlas under covering prompts, fine-

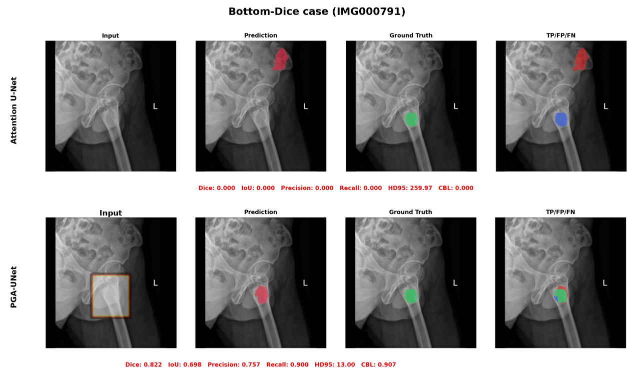


FIGURE 4. Attention U-Net (no prompt) vs PGA-UNet on BTXRD, Attention U-Net bottom-Dice case (IMG000791). Without a prompt, Attention U-Net's prediction is badly mislocated; the prompt-conditioned PGA-UNet still recovers the lesion.

TABLE 6. Comparison against promptable baselines, image-level merged, covering prompts. PGA-UNet is shown at 512 (its best configuration), 256 (resolution-matched to SAM-Med2D), and 128 (resolution-matched to ScribblePrompt-UNet); MedSAM runs at its native 1024. Raw HD95 is scored in each model's own frame, so it is not comparable across rows and is left unbolded; Dice, IoU and CBL are frame-invariant.

Dataset	Model	Res.	Dice	IoU	HD95	CBL
BTXRD	PGA-UNet	512	**0.782**	**0.661**	22.8	**0.918**
	PGA-UNet	256	0.762	0.633	10.9	0.902
	PGA-UNet	128	0.714	0.572	5.8	0.859
	MedSAM zero-shot	1024	0.304	0.185	242.4	0.832
	MedSAM fine-tuned	1024	0.724	0.590	79.9	0.883
	ScribblePrompt zero-shot	128	0.411	0.285	232.2	0.774
	ScribblePrompt fine-tuned	128	0.653	0.514	122.7	0.853
	SAM-Med2D zero-shot	256	0.255	0.156	41.2	0.734
	SAM-Med2D fine-tuned	256	0.630	0.497	17.3	0.836
FracAtlas	PGA-UNet	512	0.727	**0.585**	10.5	**0.913**
	PGA-UNet	256	0.629	0.470	10.6	0.895
	PGA-UNet	128	0.596	0.445	8.1	0.849
	MedSAM zero-shot	1024	0.319	0.193	60.1	0.843
	MedSAM fine-tuned	1024	**0.729**	0.583	16.2	0.900
	ScribblePrompt zero-shot	128	0.432	0.290	34.2	0.832
	ScribblePrompt fine-tuned	128	0.679	0.527	17.9	0.891
	SAM-Med2D zero-shot	256	0.262	0.156	21.3	0.752
	SAM-Med2D fine-tuned	256	0.563	0.417	8.0	0.846

tuned MedSAM reaches 0.729, level with PGA-UNet-512 (0.727), but only while running at a 1024 input, about four times the pixel area and 32 times the parameters; under off-center prompts PGA-UNet (0.713) pulls ahead of MedSAM (0.666) and leads on CBL in every cell. ScribblePrompt-UNet, at a parameter count close to PGA-UNet's (4.0 M vs 3.0 M), trails PGA-UNet-512 by 0.05 to 0.13 Dice and has the largest covering-to-off-center drop in the group. At matched 128 resolution, though, PGA-UNet (0.714 on BTXRD, 0.596 on FracAtlas) does not uniformly lead it: PGA-UNet is clearly ahead on BTXRD and on HD95 on both datasets (5.8 vs 122.7; 8.1 vs 17.9), but ScribblePrompt-UNet is ahead on FracAtlas Dice, IoU, and CBL (0.679 vs 0.596; 0.527 vs 0.445; 0.891 vs 0.849). PGA-UNet's margin over ScribblePrompt-UNet at 512 therefore reflects its higher operating resolution as much as its architecture, the same way MedSAM's parity with PGA-UNet above reflects a resolution and parameter advantage running the other way. The change from covering to off-center prompts is discussed separately in Section V-F.

TABLE 7. Small-lesion subset, PGA-UNet vs SAM-Med2D at the matched 256 $\times$ 256 resolution. 50 images per dataset, covering prompts, scored in a shared 512 $\times$ 512 frame.

Dataset	Model	Dice	IoU	HD95	CBL
BTXRD	SAM-Med2D zero-shot 256	0.139	0.082	62.4	0.515
	SAM-Med2D fine-tuned 256	0.256	0.170	123.8	0.561
	PGA-UNet 256	**0.705**	**0.559**	**14.7**	**0.877**
FracAtlas	SAM-Med2D zero-shot 256	0.205	0.123	33.5	0.659
	SAM-Med2D fine-tuned 256	0.371	0.256	62.1	0.736
	PGA-UNet 256	**0.648**	**0.491**	**20.6**	**0.897**

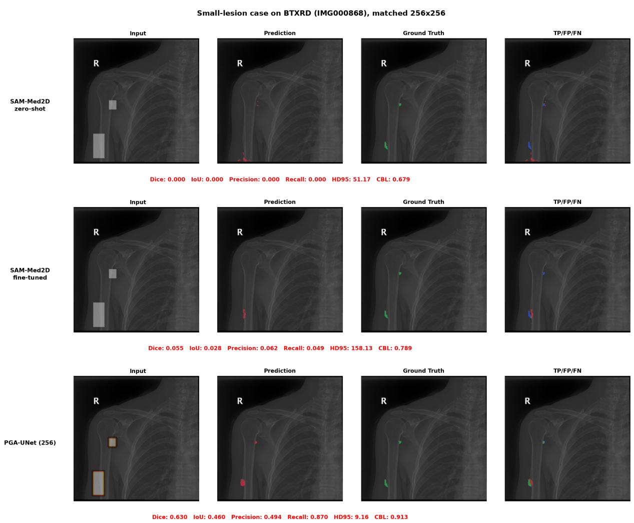


FIGURE 5. Small-lesion case on BTXRD (IMG000868, two GT polygons): SAM-Med2D zero-shot, SAM-Med2D fine-tuned, and PGA-UNet, all at 256 $\times$ 256. Both SAM-Med2D variants miss the lesions almost entirely (Dice 0.000, 0.055); PGA-UNet reaches 0.630.

E. PERFORMANCE ON SMALL LESIONS

Keeping the prompt's features active throughout the network should matter most exactly here: the small-lesion subset consists of the 50 test images per dataset with the smallest total lesion area, ranging from 7 to 433 pixels on BTXRD and 145 to 944 pixels on FracAtlas. All scores here are computed in a shared 512 $\times$ 512 frame.

a: Against SAM-Med2D at the matched resolution

SAM-Med2D struggles clearly at this lesion size (Table 7). Even after fine-tuning at 256, it reaches only 0.256 Dice on BTXRD and 0.371 on FracAtlas; in zero-shot mode it is close to nothing. PGA-UNet at the same 256 resolution still holds 0.705 and 0.648, with far lower HD95.

b: Against prompt-matched Attention U-Net at 512

Given the same box at 512, the two conventional prompt baselines trail PGA-UNet by about 0.04 on BTXRD and by 0.11 to 0.16 on FracAtlas, while prompt-free Attention U-Net drops to only about 0.28 (Table 8). The prompt-crop variant reaches a lower HD95 on BTXRD, perhaps because its tight crop suits a compact target well, but this does not carry over to FracAtlas.

TABLE 8. Small-lesion subset at 512 × 512, PGA-UNet vs Attention U-Net baselines. 50 images per dataset, covering prompts.

Dataset	Model	Dice	IoU	HD95	CBL
BTXRD	Attention U-Net (no prompt)	0.283	0.219	178.2	0.283
	Attention U-Net + prompt channel	0.728	0.591	14.2	0.879
	Attention U-Net + prompt crop	0.724	0.602	**5.5**	**0.909**
	PGA-UNet	**0.766**	**0.637**	14.0	0.897
FracAtlas	Attention U-Net (no prompt)	0.291	0.216	139.4	0.401
	Attention U-Net + prompt channel	0.610	0.449	21.9	0.888
	Attention U-Net + prompt crop	0.564	0.425	22.6	0.855
	PGA-UNet	**0.720**	**0.578**	**8.3**	**0.907**

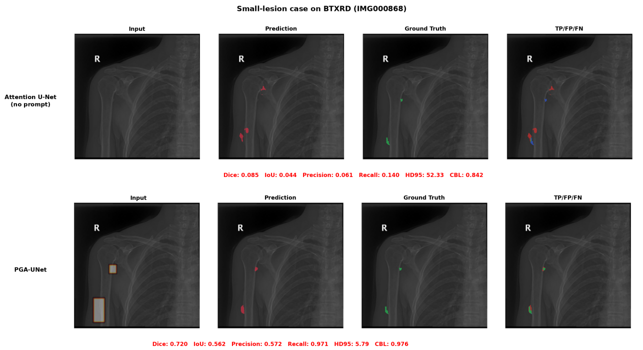


FIGURE 6. Small-lesion case on BTXRD (IMG000868, two GT polygons), Attention U-Net (no prompt) vs PGA-UNet, both at 512 × 512. Without a prompt, the automatic baseline all but misses both lesions (Dice 0.085); given the same box, PGA-UNet recovers them (Dice 0.720). The two prompt-matched Attention U-Net baselines are quantified in Table 8 but omitted here for readability.

c: Resolution on the small-lesion stems

It is exactly on this group of images that resolution shows its clearest effect. On the fixed small-lesion stems, PGA-UNet's Dice falls from 0.766 at 512 to 0.631 at 128 on BTXRD, and from 0.720 to 0.609 on FracAtlas; the loss from 512 to 128 is 0.135 and 0.111 respectively, markedly larger than the loss on the full test set (Table 9).

Overall, on precisely these smallest lesions, both prompt conditioning and input resolution matter substantially more than they do on the full test set.

F. SENSITIVITY TO PROMPT DISPLACEMENT

Table 10 shows what happens when the box is shifted off-center under one fixed rule. PGA-UNet loses only 0.008 Dice on BTXRD and 0.014 on FracAtlas, with CBL dropping by 0.006 and 0.015 respectively; on the small-lesion subset the loss stays under 0.01 for both datasets. The conventional prompt baselines lose a comparable amount or slightly more, and the three fine-tuned promptable foundation models lose 0.03 to 0.08 Dice, ScribblePrompt-UNet the most. This is only sensitivity to one displacement rule fixed in advance. We have not yet tested partial-coverage boxes or off-target boxes, so this number says nothing about boxes a clinician might freely draw in practice.

TABLE 9. PGA-UNet on the small-lesion subset across resolutions, 50 fixed stems per dataset. Dice, CBL and HD95 (native px / normalized) as covering / off-center. ′ is the covering Dice loss relative to 512.

Dataset	Res.	Dice	IoU	CBL	$HD95_{px}$ / $_n$	ΔDice
BTXRD	128	0.631 / 0.662	0.488 / 0.517	0.738 / 0.758	3.9 / 0.022	−0.135
	256	0.714 / 0.695	0.570 / 0.548	0.869 / 0.849	7.2 / 0.020	−0.052
	512	**0.766 / 0.757**	**0.637 / 0.627**	**0.897 / 0.890**	14.0 / 0.019	—
FracAtlas	128	0.609 / 0.543	0.462 / 0.393	0.828 / 0.757	8.9 / 0.049	−0.111
	256	0.642 / 0.610	0.484 / 0.451	0.887 / 0.845	10.3 / 0.029	−0.079
	512	**0.720 / 0.712**	**0.578 / 0.570**	**0.907 / 0.893**	8.3 / 0.012	—

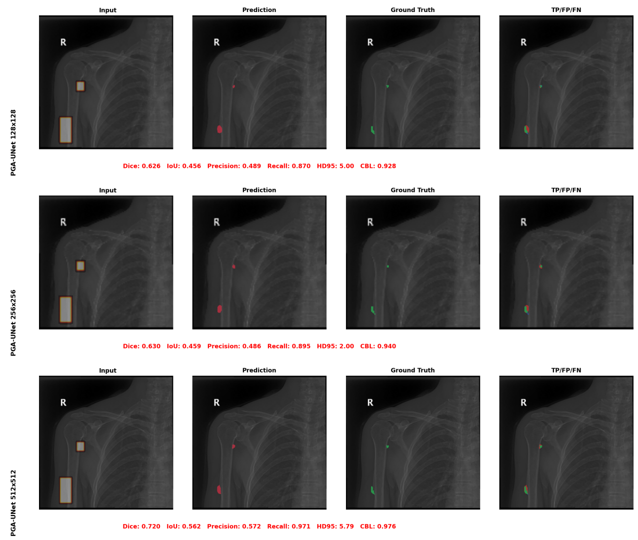


FIGURE 7. PGA-UNet on the same small-lesion case on BTXRD (IMG000868, two GT polygons) at 128 × 128, 256 × 256, and 512 × 512 (Dice 0.626, 0.630, 0.720). The prediction tightens around the lesion as resolution increases.

G. ABLATION STUDY

The ablation drops one factor at a time from the full model (Gaussian prompt combined with PSG and CAD), and separately tests replacing the Gaussian prompt with a binary box (Table 11). We do not read it as a search for the single peak configuration but for the one that stays near the top on both datasets. The full model is first or second in all four dataset-and-prompt combinations; each ablated variant is top-two in exactly one. Every setting that matches or beats it on BTXRD gives most of that back on FracAtlas: the binary-box variant leads BTXRD covering by 0.006 Dice but trails by 0.059 on FracAtlas, and PSG-only is close on BTXRD yet falls 0.092 Dice below the full model on FracAtlas. The full configuration also has the smallest BTXRD-to-FracAtlas Dice gap of any setting (0.055 vs 0.075 to 0.144 for the others). These are single-split results.

These single-split ablation gaps line up with the rest of the study. The Gaussian prompt and the CAD path cost the most to remove on FracAtlas, the harder and smaller dataset, where both drops survive Holm correction over the full family of ablation tests (Table 12); the CAD-removed variant loses 0.09 to 0.10 Dice. On BTXRD, where every configuration already sits near 0.78, nothing reaches significance, including the 0.006 Dice binary-box lead (bootstrap 95%

TABLE 10. Sensitivity to prompt displacement. Every model under covering (z) and off-center (s) prompts. Attention U-Net (no prompt) is prompt-invariant. AttUNet rows and PGA-UNet at 512 × 512; the promptable foundation models at their own native resolutions (SAM-Med2D 256, MedSAM 1024, ScribblePrompt 128), image-level merged. Raw HD95 spans these native frames and is not comparable across model families, so it is left unbolded; Dice, ´Dice, IoU and CBL are frame-invariant.

Dataset	Model	Dice z/s	ΔDice	IoU z/s	HD95 z/s	CBL z/s
BTXRD	Attention U-Net (no prompt)	0.517 / 0.517	0.000	0.418 / 0.418	120.4 / 120.4	0.620 / 0.620
	Attention U-Net + prompt channel	0.760 / 0.748	−0.012	0.635 / 0.626	33.0 / 32.6	0.895 / 0.890
	Attention U-Net + prompt crop	0.741 / 0.733	−0.008	0.615 / 0.607	24.7 / 26.4	0.912 / 0.896
	SAM-Med2D fine-tuned	0.630 / 0.601	−0.029	0.497 / 0.471	17.3 / 17.5	0.836 / 0.813
	MedSAM fine-tuned	0.724 / 0.691	−0.033	0.590 / 0.559	79.9 / 66.8	0.883 / 0.870
	ScribblePrompt fine-tuned	0.653 / 0.573	−0.080	0.514 / 0.435	122.7 / 110.5	0.853 / 0.815
	PGA-UNet	**0.782 / 0.774**	**−0.008**	**0.661 / 0.652**	22.8 / 24.5	**0.918 / 0.912**
FracAtlas	Attention U-Net (no prompt)	0.342 / 0.342	0.000	0.253 / 0.253	139.9 / 139.9	0.478 / 0.478
	Attention U-Net + prompt channel	0.593 / 0.588	**−0.005**	0.434 / 0.428	22.9 / 24.0	0.887 / 0.860
	Attention U-Net + prompt crop	0.590 / 0.563	−0.027	0.451 / 0.428	22.0 / 23.8	0.865 / 0.833
	SAM-Med2D fine-tuned	0.563 / 0.532	−0.031	0.417 / 0.389	8.0 / 8.4	0.846 / 0.816
	MedSAM fine-tuned	0.729 / 0.666	−0.063	0.583 / 0.518	16.2 / 17.7	0.900 / 0.861
	ScribblePrompt fine-tuned	0.679 / 0.598	−0.081	0.527 / 0.440	17.9 / 24.2	0.891 / 0.833
	PGA-UNet	**0.727 / 0.713**	−0.014	**0.585 / 0.570**	10.5 / 10.4	**0.913 / 0.897**

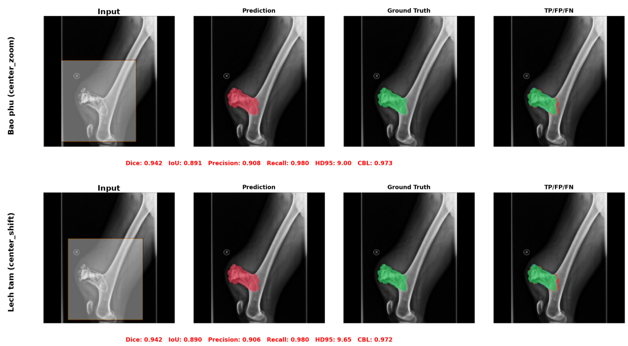


FIGURE 8. PGA-UNet on the same BTXRD case (IMG000020) under a covering prompt (top) and an off-center prompt (bottom); Dice is unchanged at 0.942.

CI $[-0.017, +0.004]$). This is the same direction seen under prompt displacement (Section V-F) and on the Attention U-Net bottom-Dice group (Table 5): the conditioning helps where the localization signal is weak and matters little where it is already strong. We therefore read the ablation as one line of converging evidence, not a component-by-component proof, which a single split could not give in any case.

H. MONTE CARLO CROSS-VALIDATION

Because a fixed split could by chance flatter or hurt the method, we retrained PGA-UNet at 512×512 on four different random image-level splits, keeping the training seed fixed at 22120196 and changing only the split seed (Table 13). The results show that Dice varies by at most 0.011 across splits, for each dataset and each prompt condition; CBL varies by at most 0.013. Notably, the repeated-split mean on FracAtlas is 0.733, slightly higher than the 0.727 obtained on the fixed split used in the article, suggesting that if anything, the split chosen for reporting is somewhat harder than typical. This is evidence that PGA-UNet is stable across different data splits, not evidence that it outperforms any baseline.

I. EFFICIENCY

PGA-UNet is very small, with all 2.955 million parameters trainable (Table 14). At the matched 256×256 resolution,

TABLE 11. Ablation at 512 × 512, image-level merged, single split. Dice, IoU, HD95 and CBL as covering / off-center.

Dataset	Configuration	Dice	IoU	HD95	CBL
BTXRD	CAD only	0.771 / 0.759	0.649 / 0.635	23.1 / 24.2	0.915 / 0.905
	PSG only	0.779 / 0.771	0.660 / 0.651	29.4 / 28.4	0.908 / 0.902
	PSG + vanilla attention gate	0.769 / 0.757	0.645 / 0.633	26.4 / 28.7	0.906 / 0.896
	Full model + binary box prompt	**0.788** / 0.770	**0.668** / 0.651	23.1 / 28.5	0.914 / 0.900
	Full PGA-UNet (Gaussian + PSG + CAD)	0.782 / **0.774**	0.661 / **0.652**	**22.8 / 24.5**	**0.918 / 0.912**
FracAtlas	CAD only	0.696 / 0.685	0.551 / 0.540	18.0 / **12.0**	0.903 / 0.885
	PSG only	0.635 / 0.611	0.479 / 0.459	14.5 / 14.8	0.894 / 0.857
	PSG + vanilla attention gate	0.690 / 0.691	0.548 / 0.547	14.0 / 14.1	0.892 / 0.881
	Full model + binary box prompt	0.668 / 0.657	0.514 / 0.505	12.0 / 13.2	0.904 / 0.886
	Full PGA-UNet (Gaussian + PSG + CAD)	**0.727 / 0.713**	**0.585 / 0.570**	**10.5** / 10.4	**0.913 / 0.897**

TABLE 12. Paired comparison of the full model against each ablated variant, per-image Dice on the single split. ´ is the mean per-image Dice difference (full minus variant), tested with a two-sided paired sign-flip randomization test (20,000 permutations). ** marks entries that stay significant ($p < 0.05$) after Holm correction over the full family of ablation tests (four metrics × two datasets × two prompt conditions); * marks raw $p < 0.05$ that does not survive correction.

ΔDice, full vs variant	BTXRD		FracAtlas	
	covering	off-center	covering	off-center
CAD only (no PSG)	$+0.010$	$+0.015^{*}$	$+0.031^{*}$	$+0.028^{*}$
PSG only (no CAD)	$+0.002$	$+0.003$	$+0.092^{**}$	$+0.102^{**}$
PSG + vanilla attention gate	$+0.012$	$+0.017^{*}$	$+0.038^{*}$	$+0.022$
Full + binary box prompt	-0.006	$+0.004$	$+0.059^{**}$	$+0.056^{**}$

it is about 92 times lighter than fine-tuned SAM-Med2D in parameter count, 12 times lighter in FLOPs, 215 times smaller in storage size, and 6 times faster on GPU. These are only cost figures measured on one specific implementation, one specific GPU, and the two resolutions used, and should not be generalized further.

J. EXPLORATORY COMPARISON OF SEGMENTATION LOSSES

Because the segmentation targets in this article are small, we tried replacing the Dice term with a size-conditioned Tversky loss, and in another trial, with a Focal Dice loss, with every other setting held fixed (Table 15). Neither option worked better: the default Dice + BCE is always best or tied on both datasets and both prompt conditions, while Focal Dice is even a clear step backward on FracAtlas, with HD95 exceeding 100 pixels. We treat this as a negative control: under the current protocol, the default objective remains the better choice, and neither of the two alternatives is adopted.

K. FAILURE MODES

PGA-UNet still makes mistakes, and these mistakes are not random but recur in a few patterns. In qualitative review, low Dice tends to cluster on lesions that are elongated or branching, lesions located in cluttered anatomy such as the shoulder or clavicle, or lesions that barely stand out from the surrounding bone. In such cases, even a good covering box does not help if the local image evidence itself is too weak.

L. TRAINING-FREE SELF-ASSESSMENT AND CANDIDATE SUGGESTION

The two support signals below are computed directly from the trained model's output and are reported as preliminary.

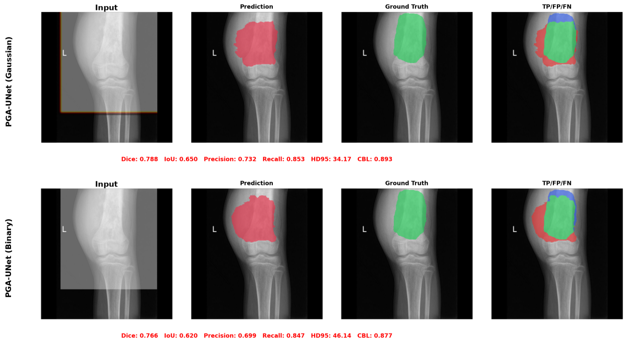


FIGURE 9. **Full PGA-UNet on the same BTXRD case (IMG000054) with a Gaussian-plateau prompt vs a binary box prompt, both at 512 $\times$ 512. The binary variant loses localization even though PSG and CAD are unchanged.**

TABLE 13. **Monte Carlo cross-validation of PGA-UNet at 512 $\times$ 512, four repeated random image-level splits, mean $\pm$ standard deviation.**

Dataset	Prompt	Dice	IoU	HD95	CBL
BTXRD	Covering	0.769 ± 0.005	0.647 ± 0.006	25.0 ± 2.5	0.905 ± 0.005
	Off-center	0.762 ± 0.004	0.639 ± 0.007	25.7 ± 2.0	0.901 ± 0.007
FracAtlas	Covering	0.733 ± 0.007	0.595 ± 0.007	14.1 ± 2.7	0.909 ± 0.006
	Off-center	0.728 ± 0.011	0.590 ± 0.010	15.6 ± 3.9	0.904 ± 0.013

Even without any ground truth, the score Q from Section III-F still tracks the true Dice of each prompt fairly closely on lesion-bearing images (Table 16). The Spearman correlation between Q and the true Dice reaches 0.70 (covering) and 0.60 (off-center) on BTXRD, and 0.65 and 0.48 on FracAtlas; all four are significantly different from zero (permutation test, $p < 0.001$), with bootstrap 95% intervals ranging from $[0.61, 0.77]$ on BTXRD covering to $[0.29, 0.63]$ on FracAtlas off-center. Both component cues contribute meaningfully: mask decisiveness alone already reaches a Spearman of nearly 0.4, jitter stability reaches about 0.5 to 0.6, and averaging the two cues is always at least as good as either alone. As a control, we also tried training a small head to directly regress the Dice; it collapsed to a near-constant output, with Spearman under 0.2 on both datasets, so it is not used.

Used in the reverse direction, Q can also rank 50 randomly sampled candidate boxes and keep only the best few (Table 17, Figure 11). For at least one of the top five boxes, 67% of BTXRD images and 82% of FracAtlas images have that box covering more than half the lesion, and the best of the five boxes covers 0.84 and 0.90 of the lesion area on average, respectively. This is only a scaffold for a small suggestion feature: Q is not a true probability, the shortlist does not replace the clinician's own review, and as the next paragraph shows, it is only meaningful when a lesion is actually present.

Q presupposes that a lesion lies inside the box. To probe this assumption, we ran the same 50-box sampling on the lesion-free images of each dataset (1879 tumor-free BTXRD images, 3366 fracture-free FracAtlas images), where no coverage is defined (Table 18). Q does not stay reliably low: the median collapses toward zero, but about 28% of boxes on BTXRD and 42% on FracAtlas still exceed $Q = 0.5$, and

TABLE 14. **Efficiency in the tested environment (NVIDIA Tesla T4, batch size 1).**

Model	Params	GFLOPs	Size	GPU ms	CPU ms
PGA-UNet 256	2.96M	7.74	11.4 MB	8.5	112
PGA-UNet 512	2.96M	30.97	11.4 MB	18.5	375
SAM-Med2D 256	271.24M	92.02	2443 MB	50.1	1324

TABLE 15. **Exploratory segmentation-loss comparison for PGA-UNet at 512 $\times$ 512, everything else fixed. Dice, IoU, CBL as covering / off-center. The default is not changed.**

Dataset	Loss	Dice	IoU	HD95	CBL
BTXRD	Dice + BCE (default)	**0.782 / 0.774**	**0.661 / 0.652**	22.8 / 24.5	**0.918 / 0.912**
	Size-conditioned Tversky	0.778 / 0.767	0.660 / 0.649	25.2 / 30.3	0.916 / 0.903
	Focal Dice	0.773 / 0.765	0.649 / 0.642	25.6 / 25.6	0.913 / 0.907
FracAtlas	Dice + BCE (default)	**0.727 / 0.713**	**0.585 / 0.570**	**10.5 / 10.4**	0.913 / **0.897**
	Size-conditioned Tversky	0.709 / 0.680	0.561 / 0.530	11.0 / 12.2	**0.918** / 0.888
	Focal Dice	0.642 / 0.578	0.493 / 0.425	105.5 / 110.1	0.802 / 0.732

almost every lesion-free image has some box placement that scores high (the per-image maximum Q averages 0.85 and 0.90). This follows from the model's weights being learned only on lesion-bearing images, with no negative class: given any box, the model still produces a decisive, jitter-stable segmentation of whatever structure lies inside, and Q rewards exactly that. Q should therefore be read as a conditional quality signal: it ranks how good a mask is once a lesion has been located, and does not by itself flag lesion absence. Adding lesion-free images to training is a direction we plan to develop.

VI. DISCUSSION

The question this article set out to answer was never simply whether a prompt is useful in general. The real question is: once a coarse box has already narrowed the search, does the way the model handles that prompt still matter? Attention U-Net without a box answers a different question altogether, and its low Dice, near 0.34 on FracAtlas, measures only how much of the task is finding the lesion. The design is tested wherever the same box is given to another model or stripped of a component: the prompt-matched Attention U-Net baselines, the fine-tuned promptable foundation models, and the ablation. Across all of them, PGA-UNet keeps a margin that is modest on BTXRD, clearer on FracAtlas, and largest on the Attention U-Net bottom-Dice group and on the smallest lesions. If the point of keeping the prompt active through the encoder and decoder is to preserve a faint lesion signal, it should help most where that signal is faintest, and that is the direction the results take: the margin over prompt-matched baselines, the per-component ablation gaps, and the robustness to prompt displacement are all largest on FracAtlas, on the smallest lesions, and under off-center prompts, and all shrink toward the easy end. The consistency of that direction across independent measurements is the main support for the design, more than any single comparison.

Beyond the main result, the study also yields two side findings. First, the loss comparison gives a fairly clear negative result: neither the size-conditioned Tversky loss nor the Focal

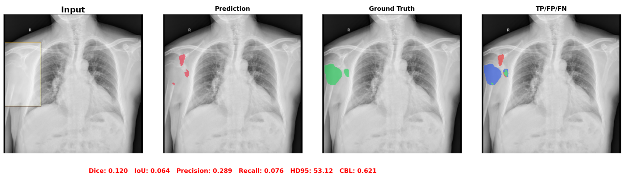


FIGURE 10. **Failure case on BTXRD (IMG000013, Dice 0.120): the prompt covers the lesion at the chest/rib margin, but PGA-UNet's prediction lands away from it. A covering box does not fix a case where the local image evidence is itself ambiguous.**

TABLE 16. **Spearman correlation between each training-free cue and the true per-prompt Dice. $Q = \mathrm{mean}(S_{\mathrm{sharp}}, S_{\mathrm{stab}})$.**

	BTXRD		FracAtlas	
Cue	cover	off-center	cover	off-center
S_{sharp} (mask decisiveness)	0.42	0.39	0.47	0.43
S_{stab} (jitter stability)	0.64	0.56	0.63	0.45
Q	**0.70**	**0.60**	**0.65**	**0.48**

Dice loss helps, and Focal Dice even makes things distinctly worse on FracAtlas, so we keep the default Dice + BCE objective. Second, the self-assessment score gives a moderate positive result: Q correlates with the true Dice at a rank level of about 0.5 to 0.7 without needing any label at all, and it succeeds where a small learned regression head does not, since that head ultimately just predicts a mean value. Even so, Q remains only a heuristic number, not yet calibrated, so the candidate list built from it should only be used to suggest where a clinician might look first, not to replace self-detection.

This study also has fairly concrete limitations. Inputs are resized to a fixed square, and the resolution results themselves show that this genuinely costs accuracy on the smallest lesions. The prompts in this article are all drawn from existing annotations, not directly drawn by a radiologist, and the protocol assumes the user has already found the correct region before drawing a box over it. Cases where the box only partially covers the lesion, is negative, or targets the wrong region are not addressed by this article; a direct check shows that the self-assessment score Q in particular stays high on many boxes drawn over lesion-free anatomy, because the model's weights were learned only on lesion-bearing images. Making Q also signal lesion absence would require training with lesion-free regions, or pairing Q with a separate lesion-presence gate. BTXRD and FracAtlas are trained completely separately, with no joint model and no cross-dataset test step, and patient-level separation cannot be confirmed from the public metadata. The Monte Carlo runs only show stability across splits, not superiority. The per-component ablation is on a single split, so it shows the direction of each component's effect but cannot settle the finer contributions; a multi-split run is future work. Finally, the gains under prompt perturbation are consistent with the design intent but have not been verified by direct feature-level analysis, so the mechanism is inferred from behavior rather than measured inside the

TABLE 17. **Candidate-region shortlist ranked by Q, keeping the 5 highest-scoring boxes out of 50 prompt-free boxes sampled per image. Coverage is the fraction of lesion pixels inside a box; full-coverage counts images whose best shortlisted box covers more than half the lesion.**

Dataset	Mean cov.	Best-of-list cov.	Full-cov. rate
BTXRD	0.52	0.84	66.8%
FracAtlas	0.62	0.90	81.9%

TABLE 18. **Distribution of the self-assessment score Q on lesion-free images, with 50 random boxes sampled per image. With no lesion present, coverage is undefined; only the distribution of Q is reported.**

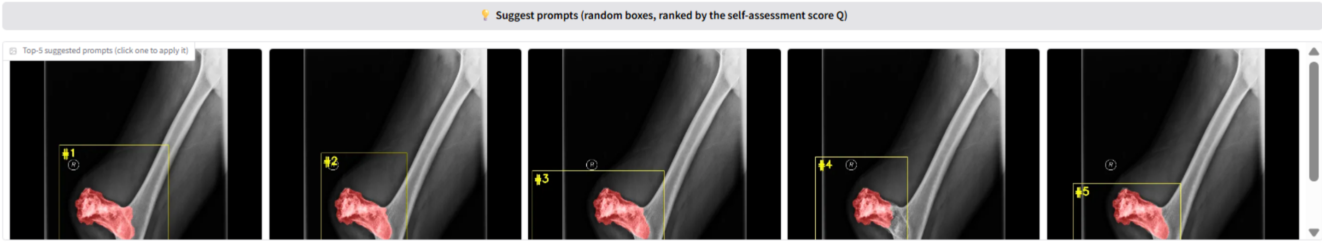
Dataset	Lesion-free img.	Median Q	p90 Q	Boxes $Q \geq 0.5$	$\overline{Q_{\max}/\mathrm{img}}$
BTXRD	1879	0.00	0.75	28%	0.85
FracAtlas	3366	0.07	0.82	42%	0.90

network.

VII. CONCLUSION

This article presented PGA-UNet, a lightweight prompt-guided CNN for interactive bone X-ray lesion segmentation. The model's design turns a user-provided lesion-covering box into a dense spatial prior, then reuses this prior through PSG on the encoder side and CAD on the decoder side. On the BTXRD and FracAtlas datasets, evaluated under both controlled covering and off-center prompts, PGA-UNet matches or surpasses the conventional prompt-matched baselines as well as three fine-tuned promptable foundation models (SAM-Med2D, MedSAM, ScribblePrompt) given the same box, with the clearest margin under off-center prompts, where automatic localization fails, and where lesions are smallest, while the model's parameter count is only about 1/92 that of SAM-Med2D and on par with the smallest of the three. Repeated-split experiments show this trend is fairly stable, and the system also exposes two label-free support signals: a self-assessment score that ranks mask reliability for boxes that do contain a lesion, and a short candidate-region shortlist, both still preliminary.

These conclusions are only valid within the protocol used: simulated boxes drawn around annotated lesions. They say nothing about unconstrained detection, off-target or partial-coverage boxes, calibrated confidence, patient-level generalization, or transfer to another dataset. In the near term, we will extend the ablation and the baseline comparisons across multiple splits, broaden the evaluation to more datasets, move toward a single shared model rather than training separately per dataset as at present, and add comparisons with more recent promptable medical models. In the longer term, we want to bring the method closer to real-world conditions: replacing the simulated boxes with looser, more realistic prompt protocols, removing the fixed-resolution resize step, and incorporating additional clinical information to improve both segmentation results and the quality of suggested regions. Finally, to reduce annotation cost and move closer to a tool usable in the clinic, the next direction is semi-supervised



Top 5 suggested prompts (checkpoint: BTXRD), ranked by the training-free self-assessment score Q. Click a thumbnail to apply it, exactly like drawing that box by hand and pressing Confirm.

Rank	Box (x1, y1, x2, y2)	CAD prompt-confidence	Q
1	(101, 196, 323, 471)	98.4%	0.96
2	(105, 212, 279, 486)	97.9%	0.95
3	(8, 248, 276, 437)	98.0%	0.93
4	(57, 219, 244, 449)	97.6%	0.92
5	(52, 274, 272, 447)	97.2%	0.91

FIGURE 11. **Candidate-region shortlist on a BTXRD case. Fifty boxes are sampled with no prompt, scored by the self-assessment score Q, and the five highest-scoring boxes are shown for the clinician to accept, adjust, or reject. A FracAtlas case is shown in the supplementary material.**

learning, exploiting the fact that drawing a coarse bounding box is much faster than drawing a pixel-accurate mask. We want to make use of a large pool of data that only has cheap box labels, combined with a small amount of accurate masks, instead of requiring a detailed mask for every sample as at present, together with turning the self-assessment score into a calibrated signal, and training with lesion-free regions so the score can also reflect lesion absence, usable for suggesting suspicious regions in an actual clinical review workflow.

DATA AVAILABILITY

BTXRD and FracAtlas are public datasets cited in this article. The manuscript reports experiments on image-level train/validation/test splits derived from those datasets. Code and split details are released at https://github.com/ThongLuc2k3/PGA_Unet2D.

AUTHOR CONTRIBUTIONS

Thong Luc: conceptualization, methodology, software, investigation, formal analysis, visualization, writing of the original draft. Nguyen Huu Binh: software support, data preparation, experiment verification, review and editing. Ly Quoc Ngoc: supervision, validation, review and editing.

ETHICS STATEMENT

This study used public de-identified radiograph datasets and did not involve prospective data collection, patient contact, or intervention.

ACKNOWLEDGMENT

This research is funded by Vietnam National University Ho Chi Minh City (VNU-HCM) under Grant number A2025-18-03.

OpenAI ChatGPT was used only for vocabulary and grammar refinement in selected non-technical manuscript text. All technical content, experimental design, mathematical formulation, figures, tables, and conclusions were prepared and verified by the authors, who take full responsibility for the final manuscript.

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